Evidence-Based Instructional Principles in Core Curricula Fraction Lessons

INTRODUCTION

Why are fractions important?

- Elementary fraction knowledge uniquely predicts students' algebra readiness and high school general math outcomes (Booth & Newton, 2012), even beyond whole number knowledge (Siegler et al., 2012)
- Fraction competence is foundational for more complex mathematics, like proportional reasoning and algebra (National Mathematics Advisory Panel [NMAP], 2008)
- Despite the importance of fraction proficiency, half of middle and high school students in the United States are not proficient with elementary fraction knowledge (NMAP, 2008)

PURPOSE

- The high percentage of general education students who are not meeting fraction proficiency standards indicates that there might be a problem with tier-one, core curriculum fraction instruction (Jitendra & Dupuis, 2015)
- The purpose of this study was to (a) systematically identify the most commonly recommended math curricula by state departments of education and (b) analyze the use of evidence-based instructional principles in core curriculum fraction lessons

METHOD

- Textbook adoption lists from states that require districts to choose math programs from predesignated lists (n=20) were systematically reviewed, and the most common four curricula were chosen for review
- Fraction lessons (n=233) from third through fifth grades were scored based on a modified instructional principle rubric originally developed by Doabler et al. (2012)

CONCLUSIONS

- This study replicates some findings from previous literature; specifically, there is a need for core math textbooks to use more explicit instruction of both concepts and procedures (Bryant et al., 2008; Doabler et al., 2012; Sood & Jitendra, 2007), but particularly fraction procedures
- Strengths of the examined math curricula included (a) the use of instructional examples of fraction concepts, (b) practice opportunities and cumulative review and (c) prerequisite skills

Examining fraction lessons in four math curricula for the use of evidence-based instructional principles revealed inadequate levels of explicit, systematic instruction.

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Table 1
Mean Grade 3 Scores for Instructional Principle Scoring Rubric

Instructional Principle	Everyday Math	Math Expressions	GO Math!
Adequate Across Curricula			
Prerequisite Skills	4.00	4.00	3.63
Instructional Examples of Fraction Concepts	3.60	3.33	3.81
Practice Opportunities & Cumulative Review	3.95	3.56	3.00
Academic Feedback	3.25	3.00	3.94
Inadequate Across Curricula			
Explicit Instruction of Fraction Procedures	2.05	1.33	1.00
Instructional Examples of Fraction Procedures	2.50	1.33	1.25
Measurement Math Models	2.25	2.78	2.38
Mixed Adequacy Across Curricula			
Math Vocabulary	3.50	2.89	2.63
Explicit Instruction of Fraction Concepts	3.20	2.44	1.00
Part-Whole Math Models	3.00	1.78	3.81
Formative Feedback Loops	2.00	2.00	4.00

Table 2
Mean Grade 4 Scores for Instructional Principle Scoring Rubric

Instructional Principle	Everyday Math	Math Expressions	EnVision Math
Adequate Across Curricula			
Prerequisite Skills	4.00	4.00	3.00
Instructional Examples of Fraction Concepts	3.24	3.26	3.20
Practice Opportunities & Cumulative Review	3.95	3.78	3.00
Inadequate Across Curricula			
Math Vocabulary	2.73	1.83	1.60
Explicit Instruction of Fraction Concepts	2.90	2.65	1.00
Explicit Instruction of Fraction Procedures	2.80	2.39	1.00
Mixed Adequacy Across Curricula			
Instructional Examples of Fraction Procedures	3.29	2.69	2.73
Measurement Math Models	2.49	3.00	3.10
Part-Whole Math Models	3.02	1.65	1.83
Academic Feedback	2.85	2.70	3.03
Formative Feedback Loops	2.12	2.00	4.00

Table 3
Mean Grade 5 Scores for Instructional Principle Scoring Rubric

Instructional Principle	EnVision Math	Math Expressions	GO Math!
Adequate Across Curricula			
Prerequisite Skills	3.00	3.95	4.00
Practice Opportunities & Cumulative Review	3.00	3.75	3.00
Inadequate Across Curricula			
Math Vocabulary	1.41	2.78	2.16
Explicit Instruction of Fraction Concepts	1.00	2.42	1.00
Explicit Instruction of Fraction Procedures	1.00	2.35	1.00
Part-Whole Math Models	2.21	1.63	2.56
Mixed Adequacy Across Curricula			
Instructional Examples of Fraction Concepts	3.28	2.93	3.12
Instructional Examples of Fraction Procedures	2.86	3.32	3.40
Measurement Math Models	3.07	2.55	2.12
Academic Feedback	3.17	2.65	4.00
Formative Feedback Loops	4.00	2.00	4.00

- The minimum acceptable criterion for each instructional principle is 3 (Doabler et al., 2012)
- While all lessons in Grades 3 through 5 were intended to be included for each curriculum, logistical limitations in accessing the curricula resulted in only three curricula per grade level